

W02 · M02 · 25 MINIT / MINUTES · Hari 1, 12:30 hingga 12:55

Hands-On SQA-AI Assistant Workshop

AI-assisted requirement review and RTM

Semakan keperluan dan RTM dibantu AI

Let the AI statically review the flawed ShopFast SRS against the BRS, rewrite weak requirements as Given/When/Then and draft the RTM as CSV.

opencode 1.18 · model Big Pickle (OpenCode Zen, free, no API key) · synthetic ShopFast data only

Bengkel SQA KRISAv2 dan PPrISA 2.0 · 14 hingga 15 September 2026 · Makmal Komputer Aras 11, Menara MyIPO

[Web guide \(copy prompts\)](#)

[Module hub](#)

[Example outputs \(GitHub\)](#)

Before you start / Sebelum mula

YOU WILL / ANDA AKAN

1. Run /review-srs on two checkout requirements
2. Compare the AI findings with your team review
3. Run /rewrite-gwt and approve the file
4. Run /rtm and open the CSV

1. Jalankan /review-srs untuk dua keperluan checkout
2. Bandingkan penemuan AI dengan semakan pasukan anda
3. Jalankan /rewrite-gwt dan luluskan fail
4. Jalankan /rtm dan buka CSV

CHECK FIRST / SEMAK DAHULU

- PowerShell is in C:\Users\<you>\sqa-krisa-bengkel\lab
- opencode starts and shows Big Pickle, OpenCode Zen (W00)

Data rules / Peraturan data

- Synthetic ShopFast data only. Never paste real agency documents or personal data.
- Free models run outside Malaysia and may learn from prompts.
- AI output is a draft. A person checks and signs off.

OUTPUT

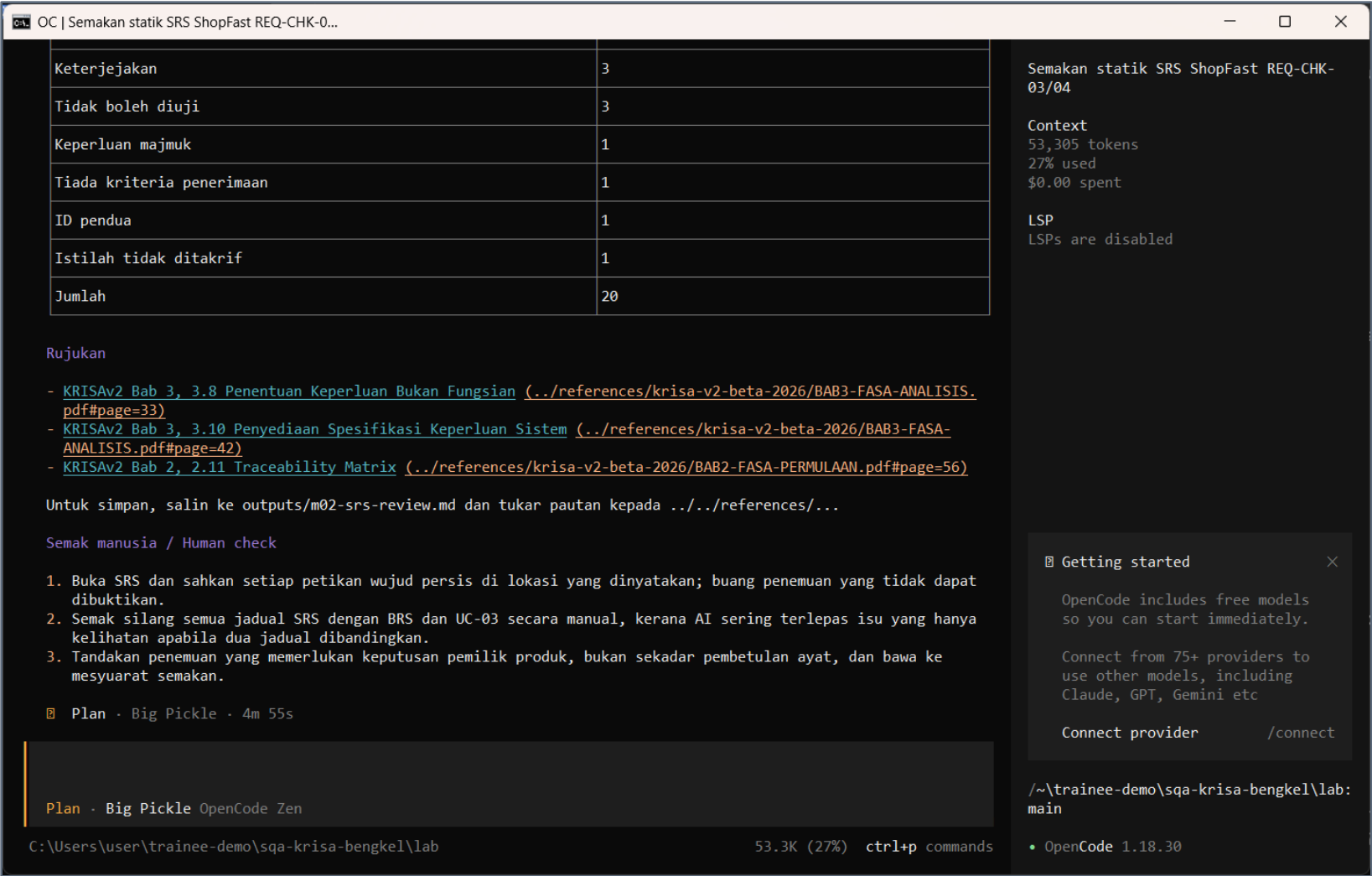
outputs/m02-gwt-rewrite.md, outputs/m02-rtm.csv

Run the SRS review command in Plan mode. It reads the SRS, BRS and RTM guide and prints findings.

Jalankan arahan semakan SRS dalam mod Plan. Ia membaca SRS, BRS dan panduan RTM dan memaparkan penemuan.

TYPE IN OPENCODE (PLAN)

```
/review-srs REQ-CHK-03 dan REQ-CHK-04
```



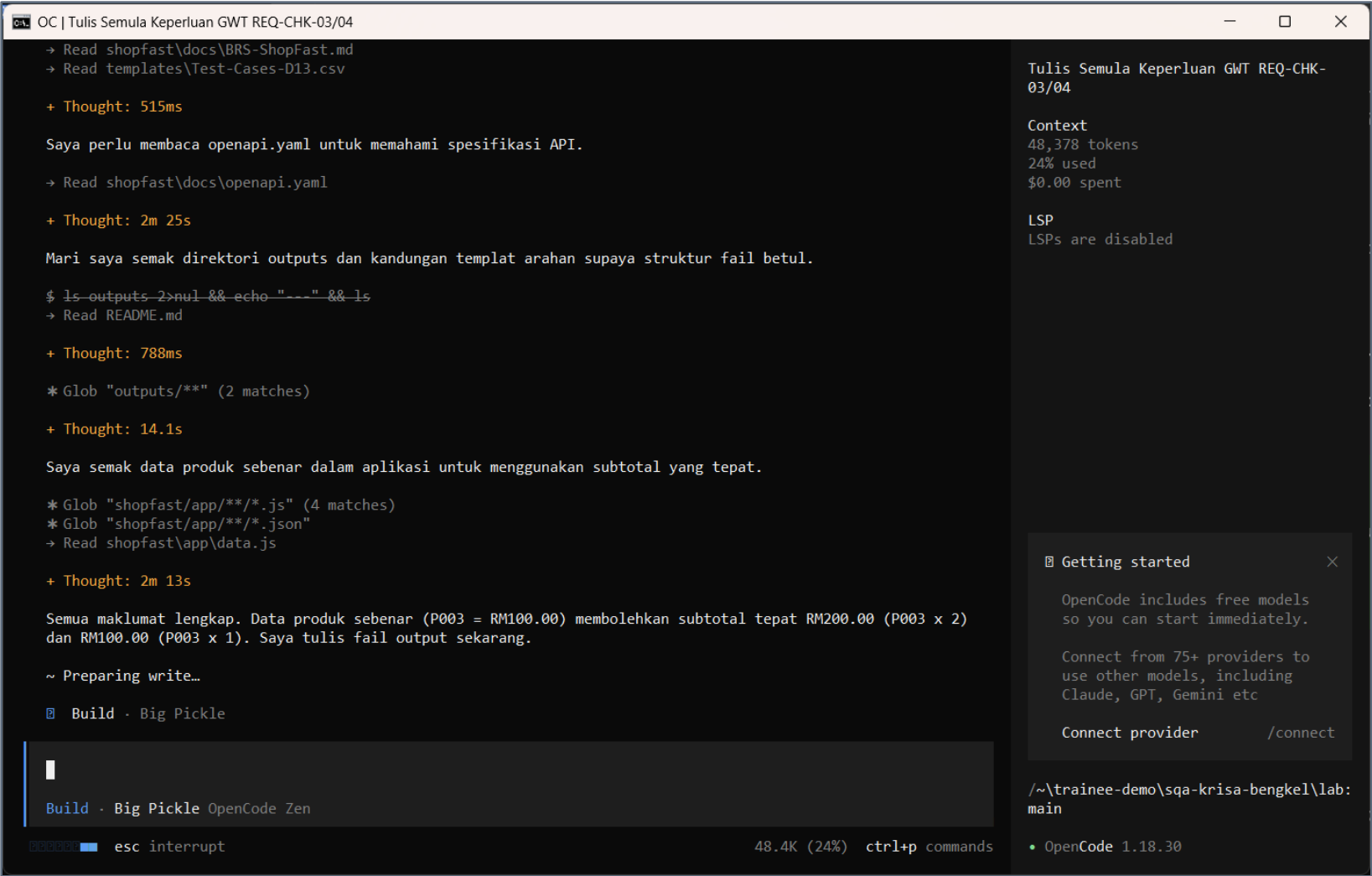
W02 · STEP 2 OF 6 · LANGKAH 2

Run /rewrite-gwt in Build mode and review the write permission. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Jalankan /rewrite-gwt dalam mod Build dan semak kebenaran menulis. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

```
/rewrite-gwt REQ-CHK-03 REQ-CHK-04
```

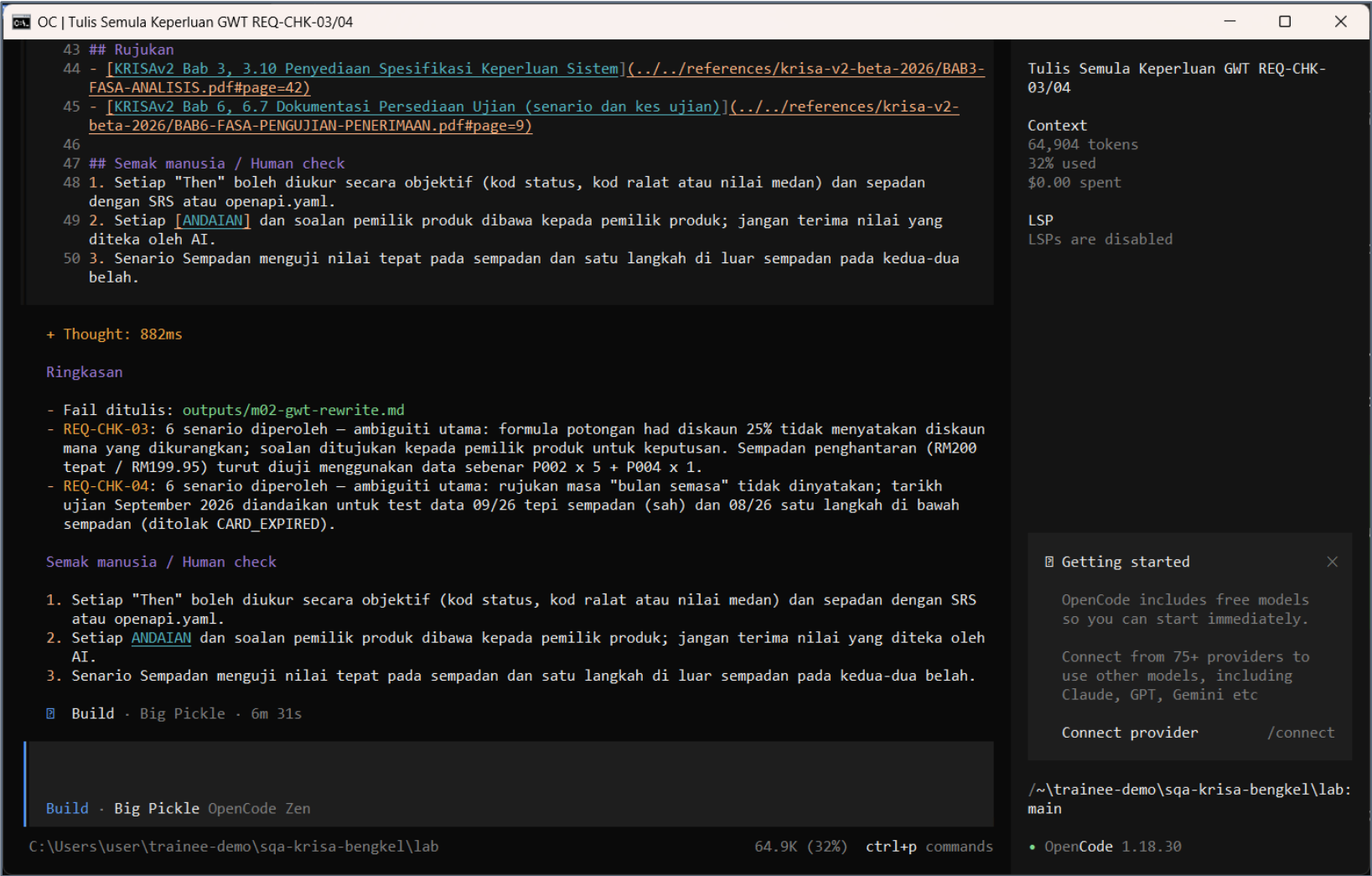


Allow once. The Given/When/Then rewrite is saved.

Allow once. Tulis semula Given/When/Then disimpan.

TYPE IN OPENCODE

Allow once (Enter)



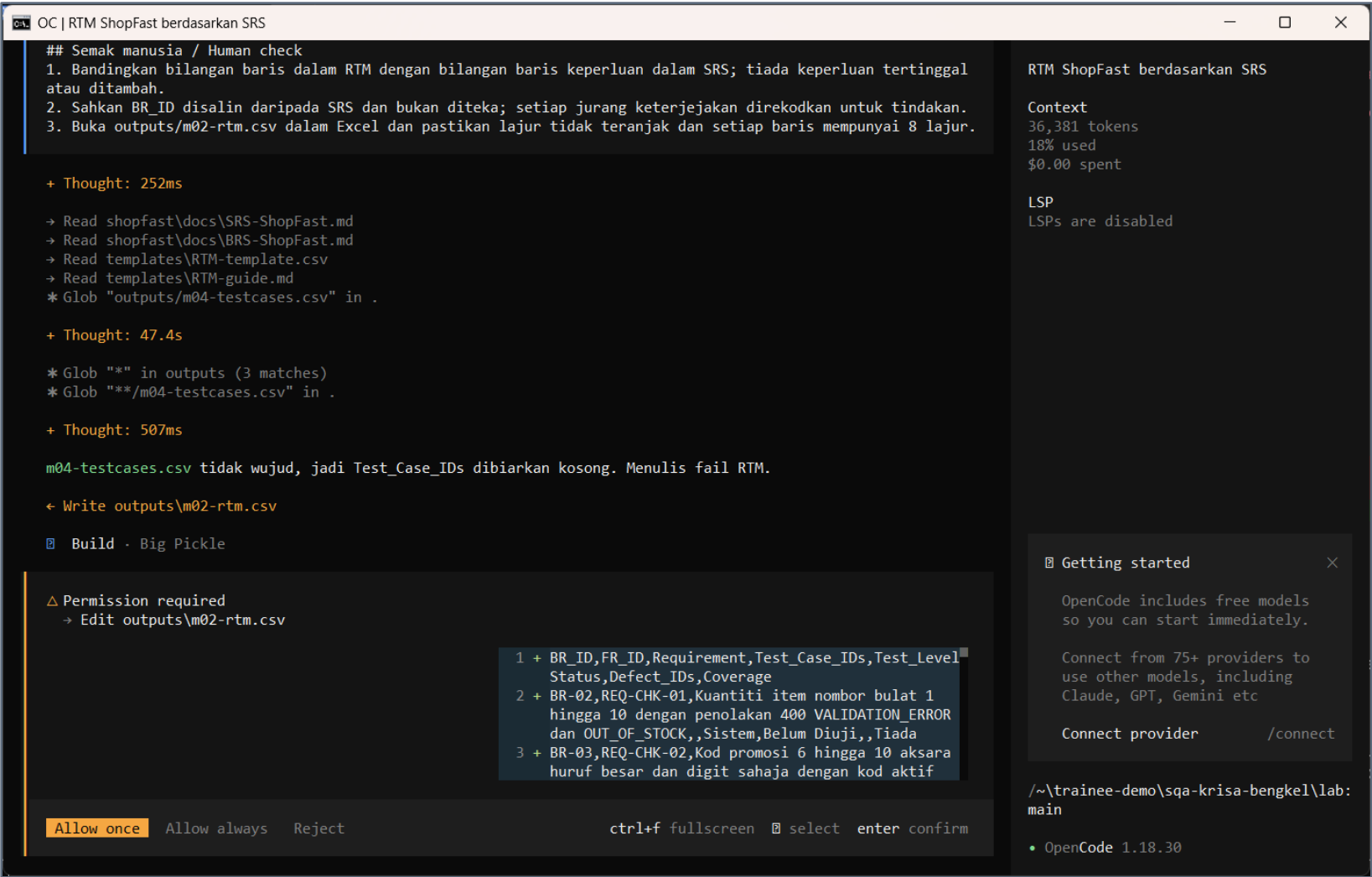
W02 · STEP 4 OF 6 · LANGKAH 4

Run /rtm to draft the traceability matrix. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Jalankan /rtm untuk mendraf matriks kebolehesanan. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

```
/rtm REQ-Chek-01 hingga REQ-Chek-06
```



Allow once. The RTM CSV is saved.

Allow once. CSV RTM disimpan.

TYPE IN OPENCODE

Allow once (Enter)

OC | RTM ShopFast berdasarkan SRS

Semakan	Keputusan	ID Terlibat
Keterjejukan ke hadapan (forward)	Gagal	BR-01; BR-07 (tiada keperluan sistem dalam skop REQ-CHK)
Keterjejukan ke belakang (backward)	Lulus	Tiada
ID unik	Lulus	Tiada
Keperluan tanpa ujian	Gagal	REQ-CHK-01; REQ-CHK-02; REQ-CHK-03; REQ-CHK-04; REQ-CHK-05; REQ-CHK-06
Ujian tanpa keperluan	Lulus	Tiada (tiada kes ujian dalam skop)
Ralat terbuka	Lulus	Tiada (tiada keperluan berstatus Gagal)

Nota: FR-05 dalam SRS (seksyen 2.5) muncul dua kali dengan maksud berbeza tetapi di luar skop RTM ini.

2. Liputan keperluan
6 keperluan, 0 Diliputi, 0 Separa, 6 Tiada. Liputan keperluan = 0 / 6 x 100 = 0.0%. Selepas m04-testcases.csv dijana, baris perlu dipetakan dengan TC_ID.

3. Rujukan
[KRISAv2 Bab 2, 2.11 Penyediaan Traceability Matrix \(RTM\), Templat T2.8](#) ([../references/krisa-v2-beta-2026/BAB2-FASA-PERMULAAN.pdf#page=56](#))

Semak manusia / Human check

1. Bandingkan bilangan baris dalam RTM dengan bilangan baris keperluan dalam SRS; tiada keperluan tertinggal atau ditambah.
2. Sahkan BR_ID disalin daripada SRS dan bukan ditekakan; setiap jurang keterjejukan direkodkan untuk tindakan.
3. Buka outputs/m02-rtm.csv dalam Excel dan pastikan lajur tidak teranjak dan setiap baris mempunyai 8 lajur.

Build · Big Pickle · 1m 53s

Build · Big Pickle OpenCode Zen

C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab38.0K (19%)ctrl+p commands

RTM ShopFast berdasarkan SRS

Context
38,050 tokens
19% used
\$0.00 spent

LSP
LSPs are disabled

Getting started

OpenCode includes free models so you can start immediately.

Connect from 75+ providers to use other models, including Claude, GPT, Gemini etc

Connect provider /connect

/~\trainee-demo\sqa-krisa-bengkel\lab: main

• OpenCode 1.18.30

Open the CSV and compare the rows with the SRS requirement IDs.

Buka CSV dan bandingkan baris dengan ID keperluan SRS.

TYPE IN POWERSHELL

```
Import-Csv outputs\m02-rtm.csv | Select-Object
-First 12 | Format-Table -AutoSize
```



Human check before you accept the output / Semakan manusia

Every finding quotes text that really exists in the SRS

Setiap penemuan memetik teks yang benar-benar wujud dalam SRS

Each Given/When/Then has one observable Then

Setiap Given/When/Then mempunyai satu Then yang boleh diperhatikan

RTM rows match the requirement IDs in the SRS, none invented

Baris RTM sepadan dengan ID keperluan dalam SRS, tiada yang direka

The AI draft becomes evidence only after this check. Record who checked it and when (Module 7 audit trail).

If something goes wrong / Jika ada masalah

opencode is not recognised

Close and reopen PowerShell so PATH refreshes. Still failing: `powershell -ExecutionPolicy Bypass -File .\setup\install-opencode.ps1 -Portable`

Scripts are disabled on this system

`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or run `opencode.cmd` instead of `opencode`

The model is slow or stops

Wait up to 2 minutes. Press Esc to interrupt, then send the prompt again. Try another free model with `/models`

Permission required dialog

Read the preview. Choose Allow once for files in `outputs/`. Reject anything outside the lab folder

File or command not found

Check the status bar path ends with `\lab`. Commands and `@files` are relative to the lab folder

Answer uses Indonesian words or dashes

Reply: Ikut AGENTS.md peraturan 1 dan 14, tulis semula dalam Bahasa Melayu Malaysia

Network or proxy error

`$env:HTTPS_PROXY = "http://proxy:port"` and `$env:NO_PROXY = "localhost,127.0.0.1"` before starting opencode

Port 3000 already in use

`$env:PORT = 3001` before npm start, and use `http://localhost:3001`

Abbreviations / Singkatan (1 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
AI	Artificial Intelligence	Software that can answer questions and write text or code by learning patterns from large amounts of data. <i>Kecerdasan Buatan. Perisian yang boleh menjawab soalan dan menulis teks atau kod berdasarkan pola daripada data yang banyak.</i>
API	Application Programming Interface	A doorway that lets one program ask another program for data or an action, without using a screen. <i>Pintu yang membolehkan satu program meminta data atau tindakan daripada program lain tanpa melalui skrin.</i>
BRS	Business Requirements Specification	The document that states what the business needs, written in business language. KRISA deliverable D02. <i>Spesifikasi Keperluan Bisnes. Dokumen keperluan dalam bahasa bisnes. Serahan KRISA D02.</i>
CSV	Comma-Separated Values	A simple table file that opens in Excel, with commas between the columns. <i>Fail jadual ringkas yang boleh dibuka dalam Excel, dengan koma di antara lajur.</i>
ID	Identifier	A unique label, such as REQ-CHK-01, so that each item can be traced. <i>Pengecam, iaitu label unik seperti REQ-CHK-01 supaya setiap item boleh dijejak.</i>
KRISA	Kejuruteraan Sistem Aplikasi Sektor Awam	The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition. <i>Buku Panduan Kejuruteraan Sistem Aplikasi Sektor Awam. KRISAv2 ialah edisi beta 2026.</i>
MyIPO	Perbadanan Harta Intelek Malaysia	Intellectual Property Corporation of Malaysia, the host of this training venue. <i>Perbadanan Harta Intelek Malaysia, tuan rumah lokasi latihan ini.</i>

Abbreviations / Singkatan (2 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
PPrISA	Panduan Pengurusan Projek ICT Sektor Awam	The Malaysian public sector guide for managing ICT projects. Codes such as PPrISA05 are its templates. <i>Panduan Pengurusan Projek ICT Sektor Awam. Kod seperti PPrISA05 ialah templatnya.</i>
REQ	Requirement	Prefix in requirement IDs such as REQ-CHK-01. CHK stands for checkout. <i>Keperluan. Awalan dalam ID keperluan seperti REQ-CHK-01. CHK bermaksud checkout.</i>
RTM	Requirements Traceability Matrix	A table linking each requirement to its design, tests and defects, so nothing is missed. <i>Matriks Kebolehkesanan Keperluan. Jadual yang memautkan setiap keperluan kepada reka bentuk, ujian dan kecacatan supaya tiada yang tertinggal.</i>
SQA	Software Quality Assurance	All the planned work that gives confidence the software meets its quality requirements. <i>Jaminan Kualiti Perisian, iaitu semua kerja terancang yang memberi keyakinan bahawa perisian memenuhi keperluan kualiti.</i>
SRS	Spesifikasi Keperluan Sistem	The System (or Software) Requirements Specification: what the system must do. KRISA deliverable D03. <i>Spesifikasi Keperluan Sistem, iaitu perkara yang mesti dilakukan oleh sistem. Serahan KRISA D03.</i>
TUI	Text User Interface	A full-screen program that runs inside a terminal window, like opencode. <i>Antara muka pengguna berasaskan teks, iaitu program skrin penuh dalam tetingkap terminal seperti opencode.</i>

Links / Pautan

[Web guide with copyable prompts](#)

[Module hub \(slides, references, templates\)](#)

[Example outputs from the real run](#)

[Lab README and command list](#)

[AGENTS.md rules used by the assistant](#)

[opencode documentation](#)

[opencode TUI commands and keys](#)

[OpenCode Zen free models and data policy](#)

[Bab 2, 2.4 Kajian Keperluan Bisnes \(BRS\) \(p.10\)](#)

[Bab 2, 2.11 Traceability Matrix \(RTM\) F1.8, Templat T2.8 \(p.56\)](#)

[Bab 3, 3.8 Keperluan Bukan Fungsian F2.4 \(p.33\)](#)